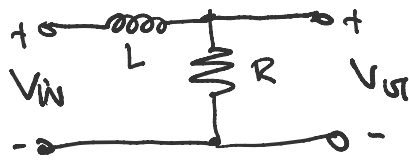


b1 a)

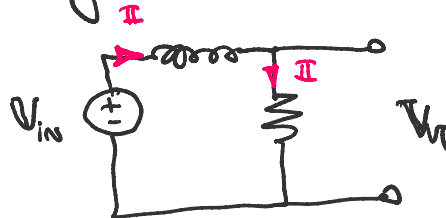


$$L = 0.5 \text{ H}$$

$$R = 200 \text{ k}\Omega$$

Sk  $B(\omega)$ .

Lösung:



KVL ger

$$-V_{in} + Z_L I + Z_R I = 0 \quad (1)$$

$$-V_{out} + Z_R I = 0 \quad (2)$$

$$(2) \Rightarrow I = \frac{V_{out}}{Z_R} \quad (3)$$

(3) i (1) ger

$$-V_{in} + (Z_L + Z_R) \frac{V_{out}}{Z_R} = 0 \quad (4)$$

MEN

$$\begin{cases} Z_R = R \\ Z_L = j\omega L \end{cases}$$

$$\Rightarrow \frac{V_{out}}{V_{in}} = \frac{R}{R + j\omega L} \quad (5)$$

Taz Snapper

$$\left| \frac{V_{out}}{V_{in}} \right| = \frac{R}{\sqrt{R^2 + \omega^2 L^2}}$$



DVS. EFT LÖFPASS-FILTER.

SWAR:

$$\frac{V_{out}}{V_{in}} = \frac{R}{R + j\omega L}$$